PSM V1 — CONNECTION-LEVEL SCHEMATIC (thesis baseline)

16x VCNL4020 on 4x4 array; 2x PCA9548A; Tiva/LaunchPad header. PCB is authoritative for passives and routing.

Per sensor: 100nF local decoupling, 10R IR-anode series resistor. 22uF bulk per mux/header.

U1



U2



U3



U4



U5



U6



U7



U8



U9



U10



U11



U12



U13



U14



U15



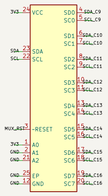
U16



U17



U18



J1



R1

R2

R3

R4

R5

R6

R7

R8

R9

R10

R11

R12

R13

R14

R15

R16

R17

R18

R19

R20

C17

C18

U17 channels 0–7 -> U1..U8; U18 channels 0–7 -> U9..U16.

VCNL4020: 1 IR anode, 2 SDA, 3 INT, 4 SCL, 5 VDD, 8/9 GND, 6/7/10 NC.

Carleton University — Instrumented Infant-Child Car Seat

Sheets: /

File: psm-v1.kicad_sch

Title: PSM V1 — Connection-Level Schematic

Size: A1 | Date: 2025-06-04 | **Rev: V1.0**

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